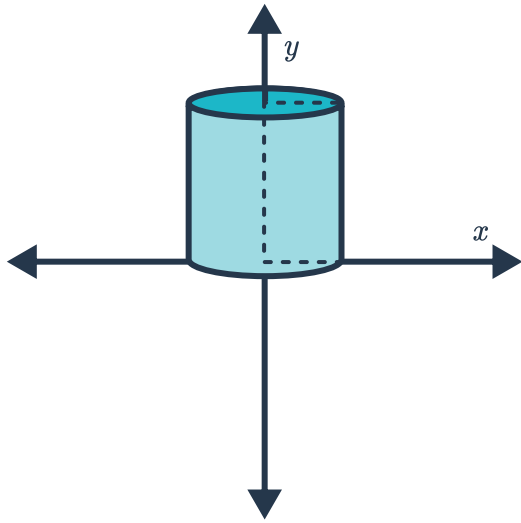


Solids of revolution

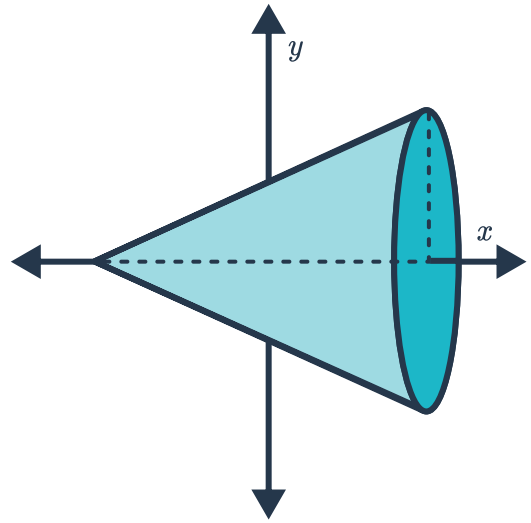


1 Identify the axis of rotation for each solid of revolution:

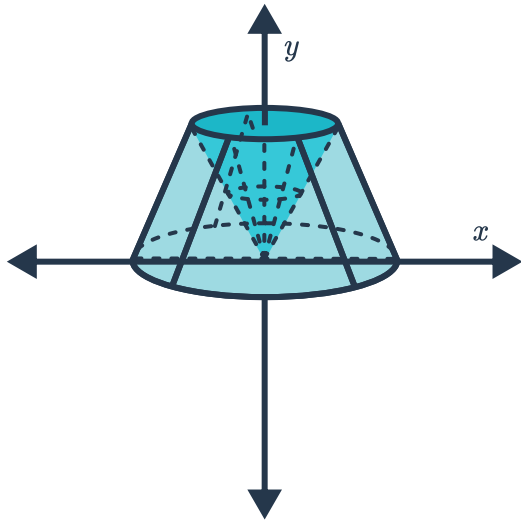
a



b



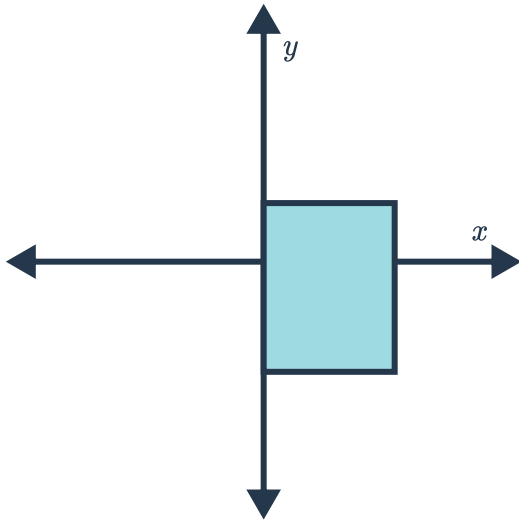
c



2 Match each figure with its solid of revolution for rotating around the y -axis:



Figure 1



Solid 1

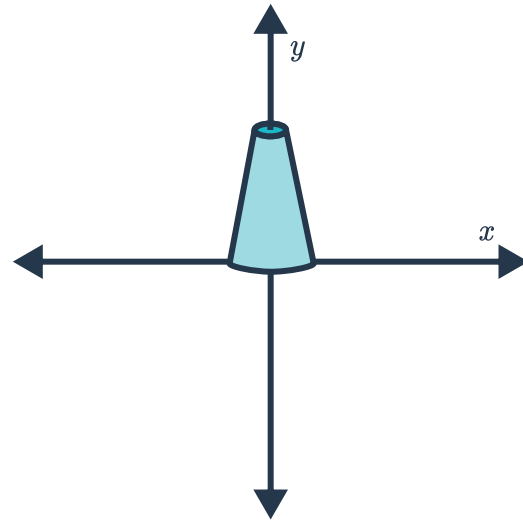
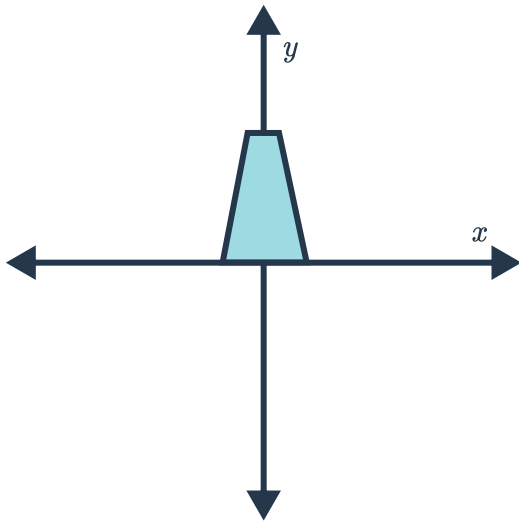


Figure 2



Solid 2

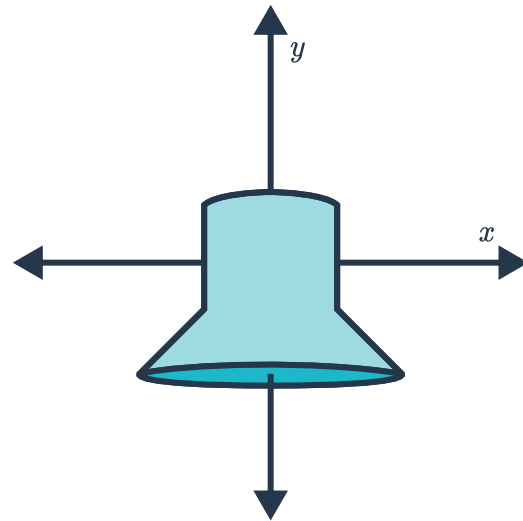
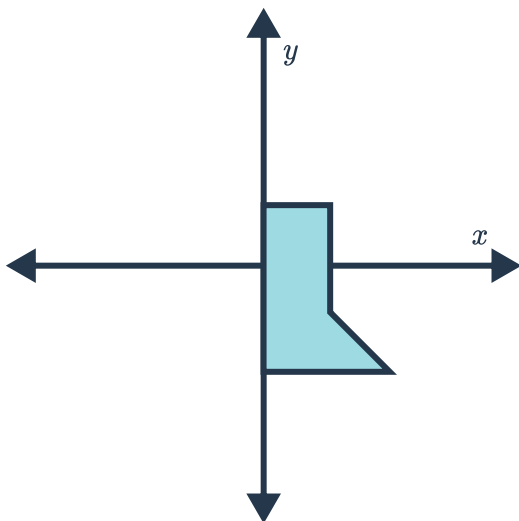
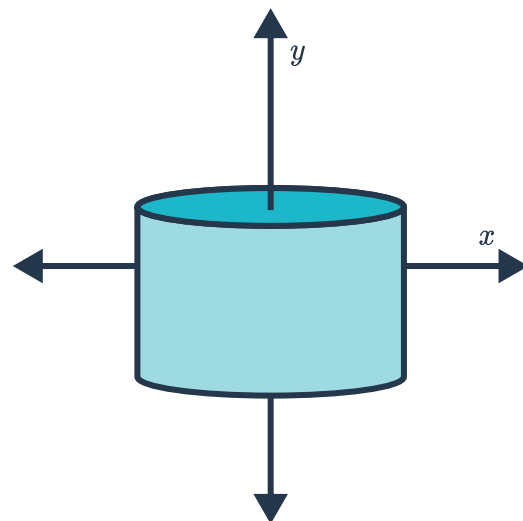


Figure 3

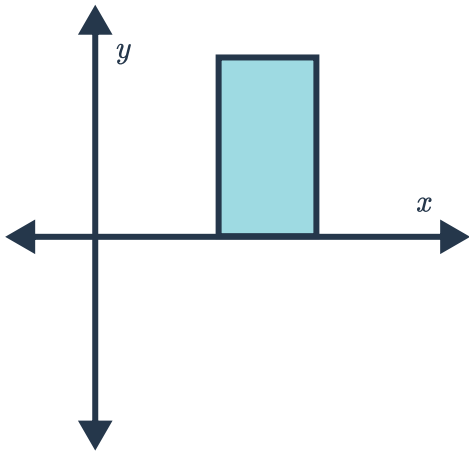


Solid 3

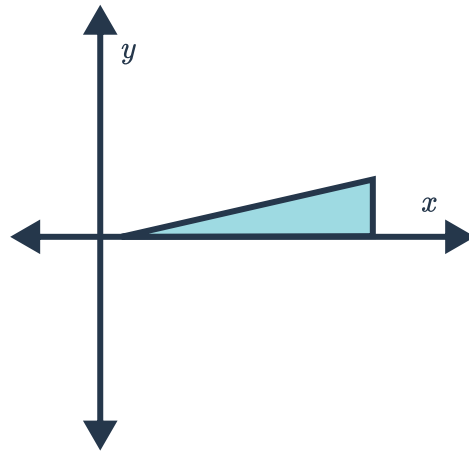


- 3 For each of the following figures, sketch the  of revolution produced by rotating around the x -axis:

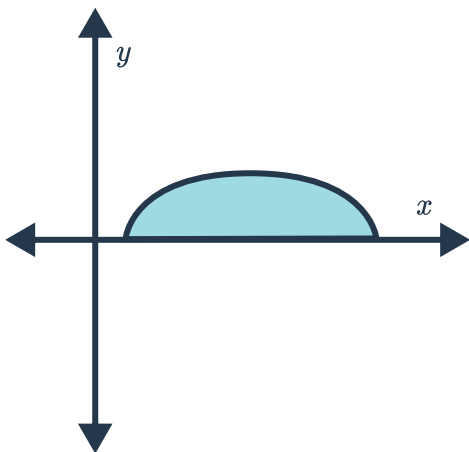
a



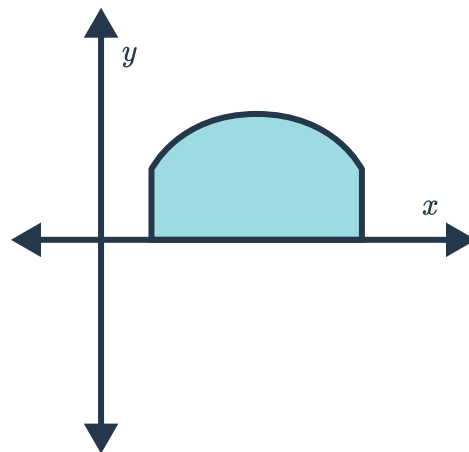
b



c

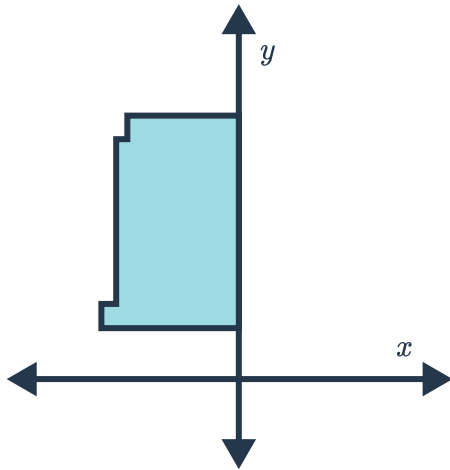


d

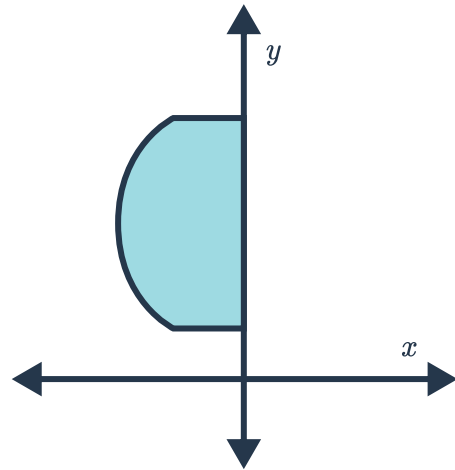


- 4 For each of the following figures, sketch the  of revolution produced by rotating around the y -axis:

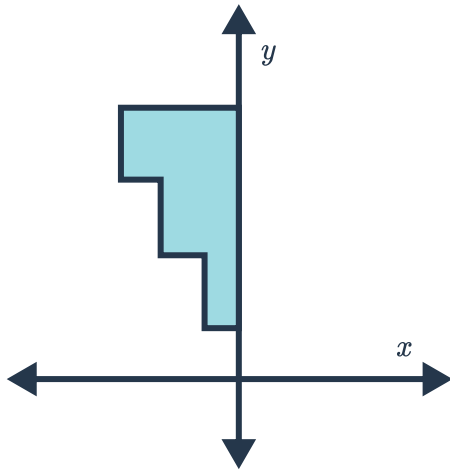
a



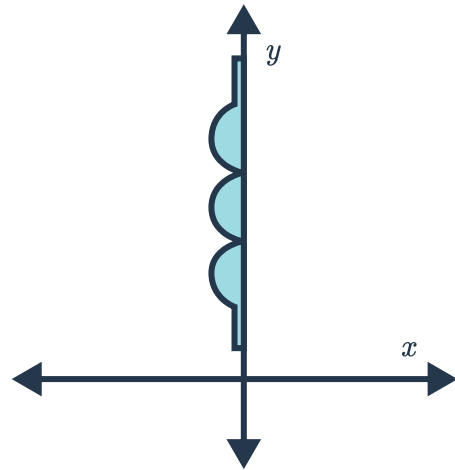
b



c



d



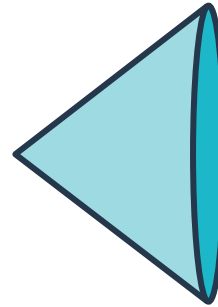
5 Sketch the two-dimensional figure that produces each solid of revolution around the x -axis:



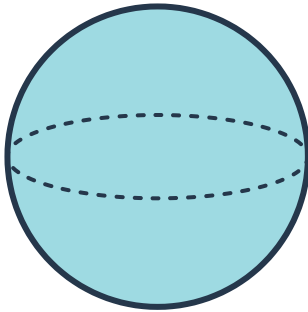
a



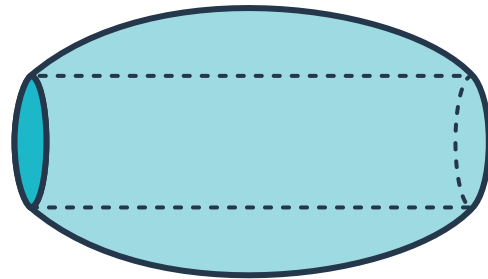
b



c



d



6

- a Sketch a two-dimensional figure on the coordinate plane that produces the same solid of revolution when rotated around the x and y -axes.
- b Sketch the solid of revolution produced by rotating your figure around the y -axis.
- c Sketch the solid of revolution produced by rotating your figure around the x -axis.

7

- a Sketch a two-dimensional figure on the coordinate plane that produces different solids of revolution when rotated around the x and y -axes.
- b Sketch the solid of revolution produced by rotating around the y -axis.
- c Sketch the solid of revolution produced by rotating around the x -axis.



Applications

8 Francesca is wanting to 3D a print bowl. She has worked out that the best way to do this is to create a two-dimensional figure and then rotate it around the y -axis.

- a Draw a template that Francesca can use to create a bowl.
- b Sketch what your bell will look like when finished.

9 A wood turner has been making a goblet (wine glass) on their lathe.

If the lathe is rotating horizontally, draw a figure that when rotated will produce the goblet.



10 Gunner works in a foundry making large bells. The easiest way to do this is by hollowing the required shape out of a block of molding clay. Gunner must construct a template that when rotated will create the hollow.

- a Design a template Gunner could use to construct the bell.
- b Sketch what your bell will look like when finished.

